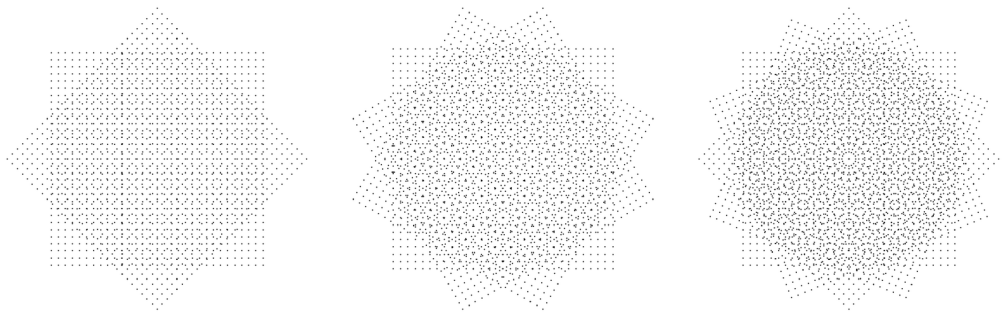
# Notes

## Chapter 10: Processes of Perception and Analysis

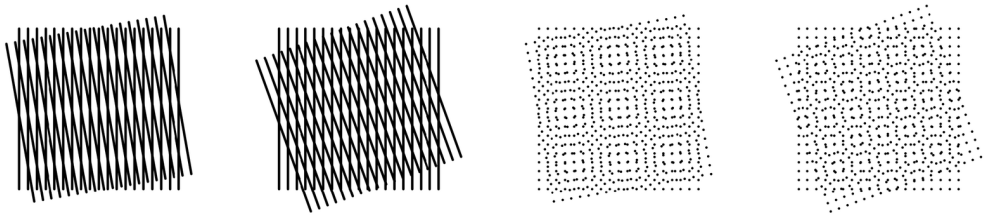
### Section 7: Visual Perception

#### Moire patterns

The pictures below show moire patterns formed by superimposing grids of points at different angles. Our visual system does not immediately perceive the grids, but instead mainly picks up features formed from local arrangements of dots. The second picture below is similar to patterns of halftone screens visible in 4-color printing under a magnifying glass.



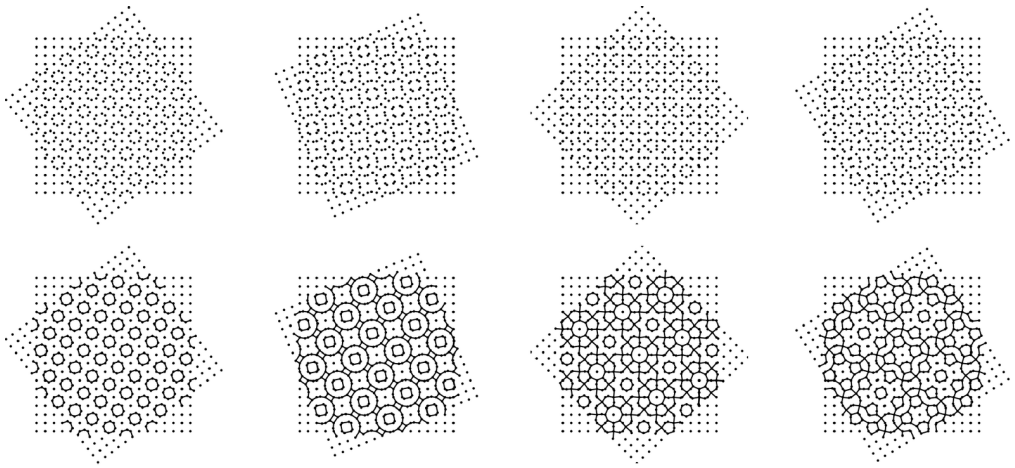
In the first two pictures below, bands with spacing $1/2\ Csc[\theta/2]$ are visible wherever lines cross. In the second two pictures there is also an apparent repetitive pattern with approximately the same repetition period.



$\theta = 10°$ $\theta = 20°$ $\theta = 10°$ $\theta = 20°$

The patterns are exactly repetitive only when $Tan[\theta] == u/v$, where $u$ and $v$ are elements of a primitive Pythagorean triple (so that $u$, $v$ and $Sqrt[u^2 + v^2]$ are all integers, and $GCD[u, v] == 1$). This occurs when $u=r^2 - s^2$, $v = 2\,r\,s$ (see page 945), and in this case the minimum displacement that leaves the whole pattern

unchanged is *{s, r}*.



3/4 5/12 20/21 28/45

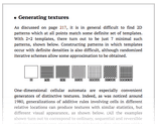
The second row of pictures illustrates what happens if points closer than distance $1/\sqrt{2}$ are joined. The results appear to capture at least some of the features picked out by our visual system.

PAGES RELATED TO THIS NOTE: 411, 582, 613

OTHER NOTES FROM PAGE 1078:

Halftoning

Perception and presentation



Generating textures

EXTERNAL RESOURCES RELATED TO THIS PAGE:

Wolfram Demonstrations Project
Moiré Patterns

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From Stephen Wolfram: *A New Kind of Science* [citation] W